

Biraj Acherjee

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[GitHub \(https://github.com/Biraj021\)](https://github.com/Biraj021) | [LinkedIn \(https://linkedin.com/in/birajacherjee\)](https://linkedin.com/in/birajacherjee) | [Portfolio \(https://birajacherjee.dev/\)](https://birajacherjee.dev/)

Education

Narula Institute of Technology, Kolkata, India
Bachelor of Technology in Electronics & Communication Engineering | **Expected Graduation: May 2028**

- **Relevant Coursework:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs), Database Management Systems (DBMS), Software Engineering Fundamentals.
- **Academic Projects:** Applied electronic systems design, digital signal processing principles.

Technical Skills

- **Languages:** C, C++, Python, JavaScript (ES6+), HTML5, CSS3, SQL
- **Frameworks & Libraries:** React, Flask, Node.js (basic), NumPy, Pandas, Matplotlib, STL (C++)
- **Databases & Cloud:** SQLite, Firebase Firestore, Firebase Authentication
- **Tools & Developer Workflows:** Git, GitHub, VS Code, Postman, Linux CLI, Vite, REST APIs, JSON

Technical Projects

Emergency Healthcare Finder | *React, Firebase, Google Maps API, Geolocation API*

- **Description:** Developed a responsive emergency healthcare locator web application enabling real-time discovery of hospitals within a local radius.
- **Key Contributions:**
 - Implemented location-based queries utilizing the **Google Maps JavaScript API** and browser **Geolocation API** to map nearby emergency services dynamically.
 - Engineered an asynchronous **SOS emergency system** that captures user coordinates and triggers immediate emergency signal updates, optimizing response workflow.
 - Structured **Firebase Firestore** databases to manage real-time hospital coordinates, establishing efficient query indexings to achieve sub-second data retrieval.

Smart Study Planner | *Python, Flask, SQLite, JavaScript, HTML5/CSS3*

- **Description:** Created an adaptive productivity platform with task scheduling, analytics tracking, and progress metrics.
- **Key Contributions:**
 - Designed a task prioritization engine modeled after the **Eisenhower Matrix algorithm**, automating scheduling priority based on task urgency and deadlines.
 - Built dynamic study analytics using front-end charting elements, enabling students to track streaks, study time allocation, and productivity indices.
 - Developed a secure user authentication system utilizing session-based login and password hashing to safeguard user planning profiles.

Banking Management System | *C++, OOP, Standard Template Library (STL), File I/O*

- **Description:** Developed a console-based banking application demonstrating advanced object-oriented design and file database management.
- **Key Contributions:**
 - Implemented core **OOP principles** (Polymorphism, Inheritance, Encapsulation) to design distinct account types (Savings, Current) and transaction classes.
 - Managed transient account memory using **STL containers** (`std::vector`, `std::map`) and ensured persistent storage by serializing account states through file streams.
 - Programmed robust exception-handling modules to validate transaction thresholds, preventing overdraft issues and invalid account state updates.

Hackathons & Achievements

- **Top 10 Finalist** | *HACKINVERSE 0.5*
 - Competed against developers nationwide to prototype and pitch a software solution under a tight 36-hour deadline. Contributed as a Core Software Developer.
- **Participant** | *HackArena & Tradition Hacks*
 - Prototyped full-stack applications within intensive 24-hour sprint environments, integrating APIs and collaborating on Git-driven codebases.
- **Best Boys Team Award** | *Synchronicity S2*
 - Recognized for strong team collaboration, technical execution, and project presentation.
- **NPTEL Certification in Database Management Systems** | *IIT Madras / Swayam*
 - Completed formal training in database theory, SQL queries, relational calculus, schema design, and normal forms.
- **Android Development Training Certification**
 - Completed structured training on Android app lifecycles, UI components, and state management.